

Criterio de la integral

1) Es decreciente?

2) Si la $\int$ converge, Σ converge

Converge si da un numero

Diverge si da $+\infty$

$$\begin{aligned} &1) \infty \\ &\Sigma_{n=2} n \cdot e^{-n} \end{aligned}$$

Decreciente?

$$f(x) = x \cdot e^{-x}$$

$$f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$$f'(x) = e^{-x} + x \cdot e^{-x} - 1$$

$$e^{f(x)} = e^{f(x)} \cdot f'(x)$$

$$= e^{-x} - x e^{-x} = 0$$

$$e^{-x}(1-x) = 0$$

~~$$e^{-x} = 0$$~~

$$1-x=0$$

$$x=1$$

$$\begin{array}{c|c|c} x & - & + \\ \hline -x & + & - \end{array}$$

$-\infty$	1	$+\infty$
$1-x$	$+$	$-$
$f'(x)$	$+$	$-$
$f(x)$	$\nearrow$	$\searrow$

$\therefore$ Decreciente

∞

$$\sum_{n=2}^{\infty} n \cdot e^{-n}$$

 $n=2$

$$f(x) = x \cdot e^{-x}$$

$$\int_2^{\infty} x \cdot e^{-x}$$

ILATE

$$\int x \cdot e^{-x}$$

$$u = x$$

$$du = dx$$

$$dv = e^{-x}$$

$$v = -e^{-x}$$

$$15. \int e^{ax} dx = \frac{e^{ax}}{a} + c$$

$$-x e^{-x} - \int -e^{-x} dx$$

$$uv - \int v du$$

$$-x e^{-x} + \int e^{-x} dx$$

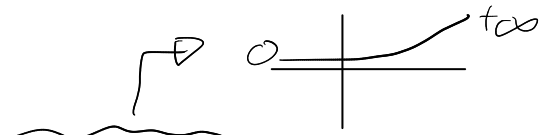
$$\left. -x e^{-x} - e^{-x} \right|_2^{\infty}$$

$$F(b) - F(a)$$

$$\lim_{x \rightarrow \infty} \left(-x e^{-x} - e^{-x} \right) - \left(-2 \cdot e^{-2} - e^{-2} \right)$$

, L'Hôpital

$$\lim_{x \rightarrow \infty} -x e^{-x} - \lim_{x \rightarrow \infty} e^{-x} - \left(-2 \cdot e^{-2} - e^{-2} \right)$$



$$\lim_{x \rightarrow +\infty} -xe^{-x} - \lim_{x \rightarrow +\infty} e^{-x} = (-2 \cdot e^{-2} - e^{-2})$$

$$\lim_{x \rightarrow +\infty} -e^{-x} + xe^{-x}$$

$$\lim_{x \rightarrow +\infty} e^{-x}(-1+x)$$

$$\lim_{x \rightarrow +\infty} \frac{-1+x}{e^x}$$

L'Hôpital

$$\lim_{x \rightarrow +\infty} \frac{-1}{e^x} = 0$$

$$-(-2 \cdot e^{-2} - e^{-2})$$

$$3e^{-2}$$

∴ Converge

∞

$$\sum_{n=1}^{\infty} \frac{n}{1+n^2}$$

$$f(x) = \frac{x}{1+x^2} \quad \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}$$

$$f'(x) = \frac{1+x^2 - x \cdot 2x}{(1+x^2)^2}$$

$$= \frac{1+x^2 - 2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{(1+x^2)^2} = 0$$

$$1-x^2 = 0$$

$$x^2 = 1$$

$$x = 1$$

$-\infty$		1		$+\infty$
$1-x^2$	+	•	—	
$f'(x)$	+		—	
$f(x)$	↗		↘	

∞

$$\sum_{n=1}^{\infty} \frac{n}{1+n^2}$$

$$f(x) = \frac{x}{1+x^2}$$

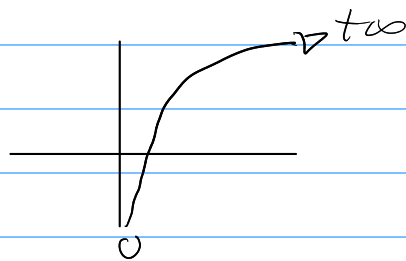
$$\int \frac{x}{1+x^2} dx \quad \begin{array}{l} u = 1+x^2 \\ du = 2x dx \\ \frac{1}{2} du = x dx \end{array}$$

$$\int \frac{1}{2} \frac{1}{u} du$$

$$\frac{1}{2} \int \frac{1}{u} du \quad \int \frac{1}{x} = \ln |x|$$

$$\frac{1}{2} \ln |u|$$

$$\frac{1}{2} \ln |1+x^2| \Big|_1^{+\infty}$$



$$\lim_{x \rightarrow +\infty} \frac{1}{2} \ln(1+x^2) - \frac{1}{2} \ln(1+1^2)$$

$+\infty - \text{finite}$

$\therefore$ Diverge

∞ $\in$ 1

$$h=3 \quad n \ln^2(h)$$

 1 $f(x)$

$$\left[\frac{1}{f(x)} \right]' = \frac{-1}{f^2(x)} f'(x)$$

$$f(x) = \frac{1}{x \ln^2(x)}$$

$$\ln^2(x) = (\ln(x))^2$$
$$2 \ln(x) \cdot \frac{1}{x}$$

$$\frac{-1}{(x \ln^2(x))^2} \cdot \ln^2(x) + x \cdot 2 \ln(x) \cdot \frac{1}{x}$$

$\therefore$ Es creciente

$$\int \frac{1}{x \ln^2(x)} dx \quad u = \ln(x)$$
$$du = \frac{1}{x} dx$$

$$\int \frac{1}{u^2}$$

$$\ln^2(x) = [\ln(x)]^2$$

$$\int u^{-2} = \frac{u^{-2+1}}{-2+1} = \frac{u^{-1}}{-1} = -\frac{1}{u}$$

$$\left. \frac{-1}{\ln(x)} \right|_3^{+\infty}$$

$$\lim_{x \rightarrow +\infty} \frac{-1}{\ln(x)} = \frac{-1}{\ln(3)}$$

$$\boxed{\frac{1}{\ln(x)} \text{ , Converge}}$$

p Series

$$\sum_{n=p}^{\infty} \frac{1}{n^p} \begin{cases} \text{Converge si } p > 1 \\ \text{Diverge si } p \leq 1 \end{cases}$$

Criterio de Comparacion directa

SOLO PARA

TRIGONOMETRICOS

$$\text{Sen}, \text{Cos} \rightarrow -1 \leq x \leq 1$$

$$\text{Sen}^2, \text{Cos}^2 \rightarrow 0 \leq x \leq 1$$

$$\sum_{n=0}^{\infty} \frac{2 + \cos(n)}{3n+5}$$

CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$-1 \leq \cos(n) \leq 1$$

$$\frac{1}{3n+5} \leq \frac{2 + \cos(n)}{3n+5} \leq \frac{3}{3n+5}$$

$$\circ \text{ IT sumadas } = 0$$

$$\circ \text{ IT multiplicadas } = 1$$

$$3+n = 0+n = n$$

$$3n = 1 \cdot n = n$$

$$\sum_{n=0}^{\infty} \frac{1}{3n+5} = \sum_{n=0}^{\infty} \frac{1}{n^2}, \text{ la serie } p \leq 1 \therefore \text{Diverge}$$

$$\frac{2 + \cos(n)}{3n + 5}, \text{ Diverge}$$

Crit de Comparacion al limite

$$\lim_{n \rightarrow +\infty} \frac{f(n)}{g(n)}, \quad \begin{aligned} & f(n) > g(n) = +\infty \\ & f(n) < g(n) = 0 \end{aligned}$$

$$\frac{n^n}{n!} = +\infty$$

$$\frac{n!}{n^n} = 0$$

CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$\text{Sean } \begin{aligned} & \infty & \infty \\ & \geq a_n & \wedge & \geq b_n \\ & n=p & & n=p \end{aligned}$$

$$\text{Con } L = \lim_{n \rightarrow +\infty} \frac{a_n}{b_n}$$

Si $L \neq 0$ (No es inf), Converge o
diverge, definido por b_n

Si $L = 0$ $\wedge$ b_n Converge,
entonces a_n Converge

Si $L = +\infty$ $\wedge$ b_n diverge,
entonces a_n diverge

CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$\sum_{n=1}^{\infty} \sqrt{\frac{n - \ln(n)}{5n^3 + 3n^2}} \rightarrow \ln(n) < n \rightarrow n - \ln(n) = n$$

$$\rightarrow 3n^2 < 5n^3 \rightarrow 5n^3 + 3n^2 = n^3$$

$$\sqrt{\frac{n}{n^3}} = \sqrt{\frac{1}{n^2}} = \frac{1}{n} \leftarrow b_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ es serie } p \leq 1$$

is Diverge

$$\lim_{n \rightarrow +\infty} \frac{\sqrt{\frac{n - \ln(n)}{5n^3 + 3n^2}}}{\frac{1}{n}}$$

$$\frac{\sqrt{\frac{n}{5n^3}}}{\frac{1}{n}}$$

$$\frac{\frac{1}{\sqrt{5}}}{\frac{1}{n}}$$

$$\lim_{n \rightarrow +\infty} \frac{n}{\sqrt{5}} = +\infty$$

is Diverge

∞

$$\sum_{n=1}^{\infty} \frac{n}{n^2+9} \approx \frac{n}{n^2} = \frac{1}{n}, \quad \text{Divergent}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n}{n^2+9}}{\frac{1}{n}}$$

$$\frac{n^2}{n^2+9} \approx \frac{\cancel{n^2}}{\cancel{n^2}} = 1$$

$$b_n \text{ divergent} \wedge \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1 \neq 0$$

$\sum_n$ Diverge

5) ∞

$$\sum_{n=1}^{\infty} \frac{2n-1}{n^3+n^2+5} \approx \frac{n}{n^3} = \frac{1}{n^2}$$

p series,

$p > 1$

Converge

$$\lim_{n \rightarrow \infty} \frac{2n-1}{n^3+n^2+5} \sim \frac{1}{n^2}$$

$$\frac{2n^3 - n^2}{n^3 + n^2 + 5}$$

$$\approx \frac{n^3}{n^3} = 1$$

b_n Converge $\wedge L \neq 0$

$\therefore a_n$ Converge